Knowledge engineering system to help understand current and historical global events

Goal:
Understand current events on a global scale.

(I have been using Claude AI to produce code)

I want to develop a "second brain" to help understand what is happening in the world.

This second brain is not "neutral", it aligns similarly to me as I curate what goes in, how it is structured and what methods are used.

HUMAN BRAIN	*KNOWLEDGE ENGINEERING SYSTEM*
Stores and finds associations between events and other information	*Stores typed, directional, sourced relationships*
Forgets where it learned something	*Enforces provenance — a claim without a source is not allowed to exist in the system.*
Treats everything it knows as roughly equally true	*Scores confidence explicitly, calibrated to source authority*
Resolves contradiction rather than holding it	*Preserves contradiction as a queryable structure*
Shaped by what arrived first, most recently, most emotionally	*Shaped by deliberate epistemological sequenc: primary voice before analyst*
Degrades under volume and speed	*Stable under volume and speed*
Cannot separate the analyst's characterisation from the primary voice	*Separates author vocabulary from knowledge base extension by design*
Drops threads when the information load exceeds capacity	*Holds thousands of grounded claims without dropping any*
Tends to construct a narrative	*Produces structure: never a story, always evidence and relationship*

Step 1: developing the foundations of the knowledge base

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Six categories of content enter the knowledge base.
Sources: Documents, speeches, articles, transcripts, legal instruments & Telegram posts

Each source carries metadata: its tier, its perspective, its author, its language, whether it has been ingested, which pipeline processed it.

Concepts: the fundamental analytical units. "Concepts" are structured claims about how things work, grounded in sources, expressed in the author's own language.

Entities: named things that appear across sources and events. People, factions, locations, resources etc.
The connective tissue that lets the system recognise that the same actor appears in a 1982 founding document and a 2024 Telegram post.

Events: significant occurrences with dates, actors, theaters, and confidence levels. Both historical events extracted from books, news articles and live events extracted from the daily feed.

Quotes: verbatim text extracted from sources, verified against the source document, linked to the concepts they ground. The evidentiary foundation everything else rests on.

I give the knowledge base a "curriculum"

Texts organised into tiers. The curriculum is an ingestion order- a deliberate sequence that encodes an epistemological commitment about which sources have authority over which claims.

Tier 0: Primary sources and doctrine from particular actors.

Tier 1: Foundational theory- numerous clusters processed in deliberate sequence which reflects logical order.
The vocabulary of the preceeding text informs and is used by the vocabulary of the second text.

Tier 2: Historical and Scholarly Analysis

Tier 3: Memoir, Oral History, Testimony, Literary

Tier 4: Contextual and Supplementary
Broader historical context, comparative cases, reference texts. Supports the higher tiers

Tier 5: Opposing doctrine: read against itself using a specific adversarial method.

Tier T: Technical papers that produce reusable analytical methodology for evaluating technical claims.

Tier L: Legal- rulings, conventions, resolutions, etc.

We want the intellectual/epistomological/relevance/chronological order to be close to the ingestion order

Hegel → Marx → Lenin → Luxemburg → Gramsci → Amin → Kadri → Roy

A revolution happens → a theorist explains why → a historian documents what actually occurred → a survivor gives testimony

The Knowledge Base consists of "Concepts"

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A concept is the analytical unit the knowledge base is built around. It is the thing that survives beyond a specific event, that can be applied to understand new events, that accumulates evidence from multiple sources over time.

Every concept has several distinct layers:

- ***The definition: written in the source author's own vocabulary.***
  Not a paraphrase nor translation into neutral academic language.
  If the author uses a specific term, the definition captures that term.

- ***The author vocabulary field: the exact term the author uses, preserved in the original language where the English translation loses precision. This is multilingual.***

- ***The knowledge base synthesis note: A synthesis relating to the concept from the pre-existing knowledge base (which can change over time).***

- ***The academic/cultural tradition or practice.***

- ***The analysis mode: strategic, symbolic, structural, technical, or legal.***
  Some concepts may span more than one mode.

How concepts relate to each other

- *Concepts are connected by typed relationships that specify not just that two things are related but how.*

- *Typed relationships:*
  ie. Grounded in/Contradicts/Extends/Exemplifies/Enriched by

- *Concept nodes are the vertices. The typed relationships are the edges.*

- *Different theoretical starting points pointing to the same phenomenon suggests significance.*

- *The knowledge bank makes connections visible and queryable in a systematic way across hundreds of concepts from dozens of theoretical traditions at once.*

The knowledge base receives information through a number of different pipelines:

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These pipelines either acquire or ingest information.

Pipelines extracts raw material and transform that material into structured records:

- *Concepts with definitions and confidence scores*
- *Events with actors and dates*
- *Quotes verified against the source text.*

And it loads those records into the database.

Ingestion pipelines: Pipelines that input raw material and outputs concepts/quotes/enrichments/cross-references/entity records

- *Book and text pipeline: the main pipeline. PDFs, EPUBs, plain text files. Handles books, monographs, reports, long-form analytical texts.*

- *Article pipeline: for scraped journalistic and analytical articles.*

- *Transcript pipeline: for YouTube and podcast transcripts.*

- *Speech pipeline: extracts declaratory claims per speech.*

- *Legal instrument pipeline*

- *Technical pipeline*

Produces: Concepts, Entities, Events, Quotes

Acquisition pipelines: getting the raw material

- *Article scraper: scrapes journalism and analytical publications.*
- *Transcript fetcher*
- *Telegram pipeline (both acquisition and ingestion)*

Entities are the connective tissue between each pipeline:

- ***Entities are named things in the world that persists across time and appears in multiple sources.***

Eg: a person, organisation, place, a resource, an operation.

Concepts accumulate across pipelines without losing their origin

- *Concepts do not belong to one source or one pipeline. It belongs to the knowledge base. When the book pipeline extracts a concept from an author, that concept enters the KB with that author as its defining source.*

- *When the article pipeline later processes a piece of journalism that touches the same concept, it writes to enrichments- adding evidence without touching the canonical definition.*

- *When the transcript pipeline extracts a speaker exemplifying the same concept, a cross-reference is generated.*

- *When the Telegram pipeline produces an event that is an instance of that structural concept, the event connects to it.*

- *Over time a single concept accumulates evidence from books, articles, speeches, transcripts, and live events.*

- *The system also can try to see what concepts it is missing, eg. if a concept uses an author's original language has vocabularly not yet in the knowledge base, this can be flagged.*

The telegram pipeline can learn from the current knowledge base which gives it context to understand groups, relevance and location.

- ***The knowledge base has historical precedents, structural analysis, the doctrine of the actors involved and theoretical frameworks that make posts from telegram understandable.***

- ***Every pipeline is always working with the most current understanding the system has produced and makes every future pipeline run more effective.***

- ***The proposals that the pipeline makes from books, articles, transcripts, speeches, legal texts, technical papers write to the same review queue before anything enters the database. Conflicts that arise get resolved here.***

Archimedes palimpsest

Every ingestion session writes a new layer over the existing structure: new concepts, new enrichments, new entity relationships, new event connections, new cross-references.

But the earlier layers are never erased.

Once the foundations are all built, a "live mode" is implemented with daily automated scrapes of telegram posts, news articles and transcripts

We want the system to have robustness and longevity

A separation of layers that cannot corrupt each other

Failures should make logs and not be silent

Audits and tests should be regularly conducted

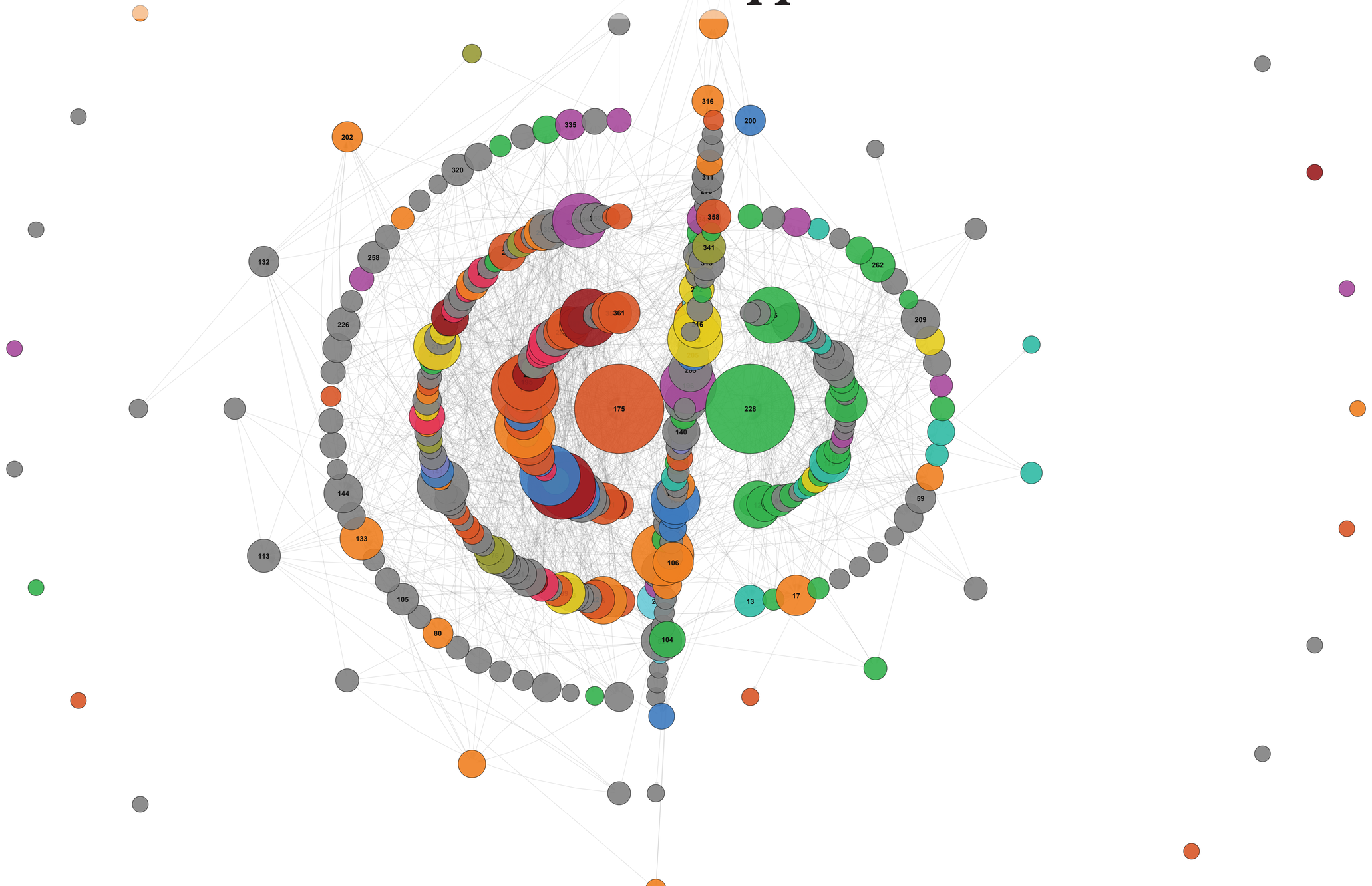
The order of intellectual sources should not "flip"

Gaps in the chain between claims and evidence are a problem

Hallucination gates at all required locations in the system!!

Hard rules that are enforced:
- *Every concept must have a source.*
- *Every reference must have a source.*
- *Confidence values of quality /validity of material affect how material is used.*
- *Every ingestion is a single transaction (no partial ingestion).*

What are some mathematical applications?



What are some mathematical applications?

Many applications in graph theory

concept graph, entity graph, provenance graph, contradiction graph, temporal graph

Graph theoretic centrality measures tell us about the importance of a concept, actor, event etc.
A structural nature of one of these types of graphs may reveal a structural nature between parties and acotrs.

Information theory

Shannon entropy, conditional entropy

Time-series analysis

Hawkes processes

We could apply this method of creating a knowledge base with pipelines to other domains.

For example:

Create a knowledge base from math textbooks ingesting:
Definitions, lemmas, propositions, theorems, corollaries.

Research paper pipeline: research papers are ingested contributing definitions, lemmas, propositions, theorems, corollaries.

Result: a queryable database that assists you in finding relevant results related to your research.